

plan to build a plant disease detection device with a camera and soil sensors

Steps to Build the Device:

1. Setup Arduino UNO

Connect your **Arduino UNO** to your PC via USB.

Install and open **Arduino IDE** to program the board.



Figure [Setup Arduino UNO]

2. Ultrasonic Sensor (HC-SR04):

Connect **Trigger** and **Echo** pins to two digital I/O pins on Arduino.

Used to measure distance between device and plants.



Figure [Ultrasonic Sensor (HC-SR04)]

3. Motion Sensor (PIR):

Connect output to a digital pin on **Arduino**.

Detects movement around the device.



Figure [Motion Sensor (PIR)]

4. DHT11 Temperature & Humidity Sensor:

Connect data pin to a digital pin.

Reads ambient **temperature** and humidity.

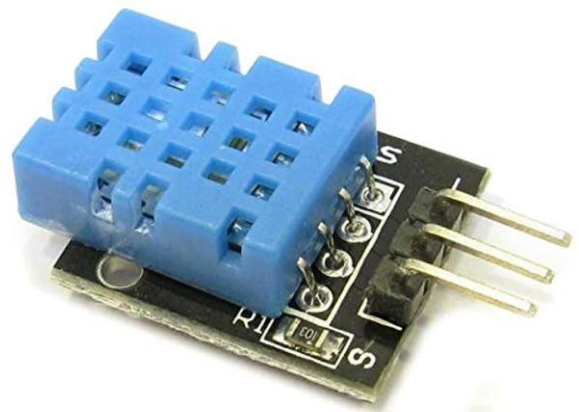


Figure [DHT11 Temperature & Humidity Sensor]

5. BME680 Sensor:

Connect via I2C pins (**SDA** to A4, **SCL** to A5 on Arduino UNO).

Measures air quality (**gas**), temperature, humidity, and pressure.



Figure [BME680 Sensor]

6. Soil Moisture Sensor:

Connect output to an **analog** input pin (A0).

Measures soil moisture levels.

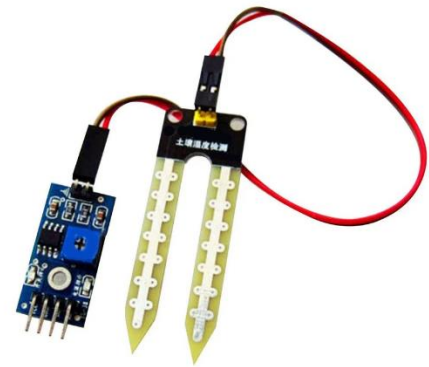


Figure [Soil Moisture Sensor]

7. LCD 2x16 with I2C:

Connect **SDA** and **SCL** pins to Arduino I2C pins.

Displays sensor **readings** in real time

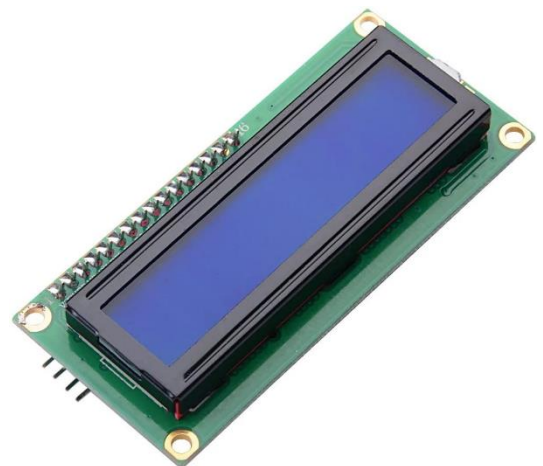


Figure [LCD 2x16 with I2C]

The measurements of your device will depend on the components and how you assemble them. Here's a rough estimate based on the components you've mentioned

Components	Approximate Sizes
1. Arduino UNO	68.6mm x 53.4mm x 15mm
2. Ultrasonic Sensor	45mm x 20mm x 15mm
3. Motion Sensor (PIR)	32mm x 24mm x 20mm
4. DHT11 Sensor	15mm x 12mm x 5mm
5. BME680 Sensor	3.3mm x 3.3mm x 1mm
6. Soil Moisture Sensor	99mm x 16mm x 4mm
7. LCD 2x16 I2C Display	80mm x 36mm x 12mm

Assembly Considerations:

- **Enclosure:** You might want to use a project box to house these components. A box measuring around ****200mm x 100mm x 50mm*** should provide enough space for all components, wiring, and allow for some ventilation.

- **Mounting:** Ensure that the sensors are mounted in a way that they can interact with the soil and environment effectively. For example, the soil moisture sensor needs to be inserted into the soil, while the camera should have a clear line of sight to the plants.

Program the Arduino

- Write an Arduino sketch that:
 - Reads sensor data from all connected sensors.
 - Calculates distance from ultrasonic sensor.
 - Detects motion with PIR sensor.
 - Reads temperature & humidity from DHT11 and BME680.
 - Reads soil moisture from analog input.
 - Displays all data on the LCD screen.